10/4/62

To: [redacted]

From:

From: SPECTRUM March-April, 1962

What we know about sleep

THE WONDER is not that we sleep but that we ever wake up. Sleeping is far less complicated than being awake, and once we understand the waking state we can say that its absence or failure is sleep.

Sleep or its equivalent is the natural condition and permanent state of the lowest forms of life and fetuses of higher forms. The human fetus, like the adult of lower forms, reacts to stimuli, and it has been known to suck its thumb. Prematures have the same electroencephalogram whether asleep or awake, and there is not much clinical difference. The infant born at term sleeps most of the time that he is not preoccupied by hunger or other painful stimuli. People who fall asleep involuntarily during or after meals perpetuate the infantile pattern.

Sleep requirement generally diminishes as age advances. When senility sets in, structural changes in the midbrain bring on drowsiness not unlike catalepsy or coma. These states, and anesthesia, alcoholic stupor and hibernation as well, differ from true sleep in that they are not easy to terminate, spontaneous neuronal activity is depressed, and oxygen uptake is decreased. There have been numerous cases of prolonged cataleptic sleep, many lasting for weeks, some for months and a few for years. They can be induced by trauma, disease or autohypnosis.

How we wake and why we sleep

It used to be thought that there was a sleep center but we now know that there is only a waking center, the reticular formation in the midbrain, which rouses us and keeps us awake. When this fails to function, or functions less efficiently, sleep supervenes. Cerebral thrombosis, encephalitis lethargica and certain drugs or poisons that affect the reticular formation can cause prolonged or permanent sleep, but there is no brain lesion that will cause permanent insomnia. The reticular formation can be deactivated from above or below. Also the primitive type of wakefulness is a subcortical affair, and the higher type, which depends upon discrimination, involves the cortex. Electric stimulation of the reticular formation or a dose of amphetamine produces nondiscriminatory wakefulness. The cortex is unable to control the reticular formation in such circumstances.

Pavlov evolved the theory that sleep is equivalent to inhibition, as encountered in the modification of a conditioned reflex. His animals went to sleep when they were subjected to rhythmic and monotonous stimuli and when an expected reward was delayed or withheld.

Rhythmic impulses induce sleep in humans too. An extraordinary example of this was seen in a heroic group of Oxford students who had their eyes glued open with collodion so that they were forced to see blinding flashes of light while listening to deafening dance music and submitting to violent electric shocks, all in rhythm. Despite the pain and inconvenience of the experience, the rhythm put them to sleep by a sort of autohypnosis. The stimuli, coming with predictable regularity, contained nothing for the cortex to come to grips with. When monotony creates boredom the cortex tells the reticular formation to stop bothering it. In the same fashion a rabbit hypnotized by a snake not only succumbs to the rhythm of the snake's movements but becomes lethargic because it remains motionless in order to play dead. This is monotonous even in the presence of grave danger.

To return to Pavlov's animals, they would probably have gone to sleep sooner without any stimulus whatever. Normal cats will sleep 18 to 20 hours a day if kept in a dark, quiet room.

Monotony may also cause irritation or depression, and insomnia may result. And certain rhythms of flicker produce convulsions in some people.

The significance attached by the sleeper to different stimuli is an all-important factor. The ticking of a clock, the drip of a faucet or the noise of the road may soothe one to sleep by their very monotony, or they may keep one annoyed if they arouse feelings of irritation or anxiety.

When a fresh stimulus impinges on the brain during sleep, the reticular formation transmits it to the cortex, where its significance is weighed. Any threat, such as pain or an ominous sound, alerts us at once, but if the cortex judges the stimulus to be unimportant, we may simply grunt or roll over, our eyes move rapidly as in light sleep, the lids may flutter momentarily, and the episode usually passes into oblivion. The sensory cortex rests during sleep but is always at the ready.

Sleep also results when neuromuscular fatigue curtails afferent stimuli. Left to its own devices, the reticular formation seems to close down for the night.

Parasympathetic stimulation causes fatigue and drowsiness, which mimic some of the physiologic effects of sleep, including peripheral vasodilation, contraction of the pupils, increased sweating, and diminished respiration, pulse rate, blood pressure and muscle tone. Intestinal peristalsis and gastric secretion decrease in sleep (though the reverse is true in people who have or are going to have a peptic ulcer). Because respiration is slowed, blood concentration of carbon dioxide rises. By testing any group for hypercapnia, it is conceivably possible to predict who will first fall asleep.

Neuronal activity is not so much diminished as altered in sleep. Spontaneous discharge of impulses from the reticular formation may continue unabated but becomes synchronized rather than irregular. There appears to be a balance in the intensity and frequency of excitatory and inhibitory impulses from cell to cell. A man can respond to his name whispered while he is in moderately deep sleep, and yet not awaken. Subjects can be taught to press a button to avoid rhythmic electric shock even in deep sleep. Claims are made that complicated lessons can be learned during deep sleep, but without much foundation. It is possible for a man to learn from a record during the predormitum, when he is highly suggestible, but not when he is sleeping soundly. Conceivably animals would make better pupils than humans because they seldom fall into the very profound sleep that is a product of our sheltered civilization.

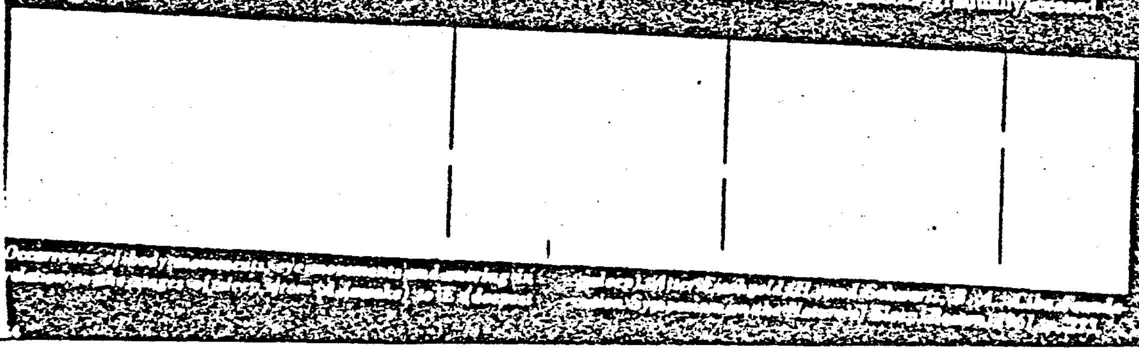
The physiology of dreams

Dreams occur when we pass from deep sleep to light sleep. They are stories that we tell ourselves, conceivably in order not to wake up completely, and we are usually so interested that it takes a considerable stimulus to pull us out of them. Rapid eye movements accompany this stage of sleep, and they are used for scanning the dream, although the eyes are primarily obeying a primitive instinct to be on the lookout for danger. Newborn infants have the same movements but their electroencephalographic pattern indicates that they do not dream. It has been suggested that the scanning and the dream may be in response to secretion of epinephrine, whose structure is basically that of the hallucinogens. The illusions that partake of true hallucination, however, are distinct from dreams; they occur when we are falling asleep rather than when we are waking up, and we are aware of their unreality. Body movements sometimes reflect the subject matter of dreams but are more likely to occur between them. W. C. Dement (Mt. Sinai Hospital, New York) has found that subjects habitually awakened at the start of their dreams become frustrated and irritable. Then when they are permitted to dream in peace, they spend much more time dreaming than they did in preliminary observations. Dreams that are not nightmares probably help us keep our emotional balance. Most people have at least three or four dreams in a night and there is often continuity from one to another.

Sleepwalking is an extension and an acting out of dreams, and is also a sign of emotional disturbance, frequent in sensitive, anxious children. Purposeful and compulsive but senseless acts may be performed as though the cortex were asleep and incapable of judgment. For example, a dreamer may carry a flashlight in a lighted room and cling to it with tenacity, refusing to accept a substitute; or if he is willing to relinquish it, a bottle may tempt him just as well. If the eyes are open, the pupils are usually dilated but react to light. The sleepwalker may retain a vague recollection of the episode.

Somnambulism has been provoked in children by giving them large doses of sodium chloride at bedtime. This may bear out the theory that neuronal activity is stimulated by sodium: when neurones are at rest they appear to have given up sodium and taken up potassium. The experiment does not take into account the diuretic action of sodium salts, which may have played a part in disturbing the children's sleep.

A man who habitually attacked his wife in his sleep was cured by psychotherapy. As he had been resentful first of his mother and later of his wife, he was advised to adopt a more assertive attitude toward his mother, and the somnambulistic episodes gradually ceased.



Occurrence of body movements, eye movements and dreaming in [illegible]

Uncomplicated narcolepsy may be merely an exaggeration of the diurnal drowsiness pattern of normal individuals. As the drowsiness preceding the sleep attack is usually prolonged, a fair proportion of narcoleptics have visual and auditory hallucinations before blacking out. The condition never remits, never seems to get much worse, appears to have a familial tendency and may be ascribable to lapses of the reticular formation. Narcolepsy with cataplexy is usually precipitated by laughter, sometimes by anger or surprise. Here again there is a parallel in normal behavior, for we sometimes become weak with laughter and many people regularly feel weak when falling asleep. Sleep paralysis, related to narcolepsy, may accompany hallucinations before sleep, or dreams at the time of waking. Psychogenic sleep, precipitated by psychic trauma, may partake of some aspects of sleep paralysis. Hypnosis is a transitional state between waking and sleep, and often leads to sleep. The knee jerk, present in hypnosis, is absent in sleep.

Snoring is due to flaccidity of the oropharyngeal muscles and may be aggravated by advancing years when muscle tone decreases further. It is rare in the light sleep of dream sequences and is most incessant in the stages of deepest sleep.

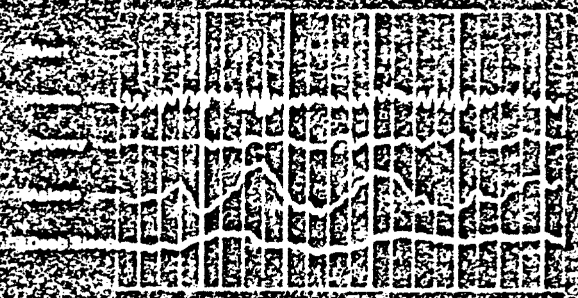
The yawn, commonly associated with oxygen hunger, is primarily a method of combating loss of muscle tone in inactivity and boredom. Many muscles participate and many more when there is concomitant stretching, as is usual in children and animals. Adults yawn most often when about to drowse off and stretch when they wake. There may be a yawn center in the medulla oblongata.

Most people experience sudden, uncoordinated bodily jerks in light sleep of short duration. When this is constantly repeated, the question has been raised whether the phenomenon is not an epileptic equivalent.

Prolongation and curtailment of sleep

Sleep therapy is popular in continental Europe and particularly in Russia where it is prescribed with the purpose of resting and restoring the cortex. Good results are reported in anxiety neuroses, reactive depressions and many other conditions. By administration of tranquilizers and barbiturates, patients are kept unconscious for 12 hours a day or more, and even around the clock, for as long as 2 or 3 weeks.

The tradition that young adults should have eight hours of sleep was pretty well confirmed by a British expedition to North Greenland lasting two years. The average time of sleep was 7.9 hours and it was little influenced by the great variation in the hour of going to bed in continuous light or dark. Winter insomnia was compensated for by naps. Of course many people can get along with considerably less sleep, but extremes should probably be discounted. Thomas Edison, for example, was reputed to require not more than four hours in bed, but he made up for it with naps on strategically placed [illegible]. Napoleon, who boasted that he needed very little sleep, dozed in the saddle after virtually sleepless nights.



Electroencephalogram patterns in waking and sleep. After Brazier, M.A.B.: The Electrical Activity of the Nervous System. New York, 1960, p. 244.

It is possible to vary both the hour and duration of sleep periods but some physiological adjustment is necessary, presumably involving electrolyte balance in the brain cells. Nathaniel Kleitman found that certain young subjects could adapt to either a 21-hour or a 28-hour day, whereas others could not. On Arctic expeditions it has been found that most men are not able to make such adjustments and get depressed when they try. Workers who alternate between day and night shifts become exhausted earlier than those who keep regular hours.

Two men competing for a prize managed to stay awake for 168 hours and 33 minutes, at which time the contest was terminated. From the fourth day on they had memory lapses, hallucinations and paranoid reactions. They were quite disorganized, experienced *déjà vu* and at one time *folie à deux*, and suffered from ankle edema. The disorganization and memory lapses persisted, but not the hallucinations; one of the subjects was hospitalized for four months after the experience. Such disruption of the personality accounts for many confessions, often false, elicited by the police. The record established by these men has since been broken by a pair of disk jockeys. And there is an unauthenticated report that a man stayed awake for 131 days in the year 1895. He expected to be able to go without sleep indefinitely but presumably did not succeed.

We Americans spend millions on soporifics. For those of us who cannot throw off anxiety and restlessness by counting sheep, there is assuredly no harm in resorting to a tranquilizer or a sedative. One of the medium-acting barbiturates, one capsule daily under continued supervision, is not likely to lead to addiction. Chloral hydrate is a satisfactory substitute. The strong-willed can fall asleep by thinking of nothing or concentrating on the rhythm of respiration. The self-deluded may successfully dispense with drugs by following compulsive routines such as obeying the folklore superstition that sleep comes to those who lie facing south or toward a church.

Though sleep be the [illegible] of death it is [illegible] better than medicine to cure the ills of the waking state. The cerebral cortex requires its normal complement of sleep in order to function properly.

Sea to Sea

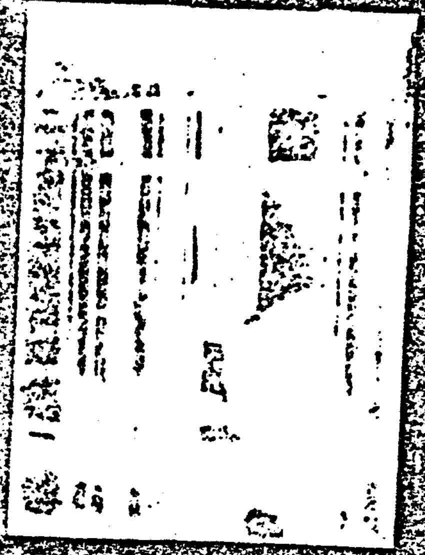
Some 300 million years ago, a few hardy aquatic creatures tentatively crawled up onto dry land in search of a new food supply. Some stayed, and over the years evolved into air-breathing animals, mammals and, finally, man. Although this momentous step was the key one in the evolutionary process, an obvious by-product was that most mammals lost the talent for existing under water.

At Holland's respected University of Leiden, however, a team of Dutch scientists has succeeded in partially reversing this [illegible] did it by [illegible] possible [illegible] of keeping [illegible] dogs who were completely submerged in water for up to eighteen hours. Not only were the animals able to breathe the water, but the dogs readapted to air-breathing without so much as a hiccup—suggesting that man may someday be able to go back where he came from.

According to 37-year-old Dutch physiologist Dr. Johannes Kylstra, the Leiden researchers first had to decide why animals drown. "Lack of oxygen is, of course, the obvious cause of drowning," Kylstra explained last week. "But equally important are the damaging effects of fresh water on lung tissue and the fatal dilution of the blood's salt content." To [illegible] drown-proof [illegible] the experimenters first applied the scientific law which [illegible] the solubility of oxygen in water. In a room-size atmospheric-pressure chamber, they oxygenated [illegible] of ordinary tap water, by [illegible] to 3117.6 pounds per square inch, which [illegible] than at [illegible]. Next, they [illegible] with salt until it had about one-third the salinity of sea water. This [illegible] because salt [illegible] chemically [illegible] the fluid [illegible] the human [illegible] (this fluid itself [illegible] a vestige of man's aqueous beginnings).

Then seventeen white mice were placed in the tank. Although the rodents' lungs quickly filled with fluid, their respiration seemed normal. One mouse, in fact, [illegible] about for a full [illegible] hours (photo) before settling down to this aquatic existence. But the experiment was only a limited success, since the mice were too small for the researchers to drain their lungs, and all of them died shortly after return to the air.

When dogs were used, however, Kylstra's colleagues were able to drain their lungs with a suction [illegible] preparatory to [illegible]. The dogs then were quickly removed from the tank and their lungs were repeatedly inflated with oxygen. After a few minutes, Kylstra [illegible]



Mouse underwater: Breathe deep.

reports, "the dogs resumed breathing spontaneously [illegible]. One month later they have shown no ill effects from their dunking."

What remains a mystery, however, is how the animals extracted oxygen from the water. Kylstra theorizes that their lungs were able to absorb oxygen through tiny capillaries the same way the lungs get oxygen from the air. [illegible] in the U.S. [illegible] that similar to Kylstra's [illegible] conducted at Stanford University's School of Medicine by Dr. Robert Goodlin, who is trying to develop an artificial uterus for [illegible]. Goodlin has [illegible] human [illegible] weeks old and kept them [illegible] for up to 22 hours in a highly [illegible] solution [illegible] through the [illegible]

Aside from pointing up the potential [illegible] of Goodlin's artificial womb concept, the Dutch achievement may have other important applications. "Our experiments proved," Kylstra says, "that the [illegible] of [illegible] after submersion in salt water are [illegible] as high as in fresh water." Thus the number of [illegible] drownings, he believes, might be sharply reduced by adding salt to the water. And if [illegible] however [illegible] human beings may be able to [illegible] from [illegible]